

Unit 11: Classification

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Summary

Keywords

Self Assessment

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Objectives

After this Unit, you will be able to

- Understand the concept of supervised learning.
- Learn the concept of classification and various methods used for classification.
- Know the basic concept of binary classification.
- Understand the working of naïve Bayes classifier.
- Analyze the use and working of Association based and Rule-based classification.
- Know the working of the KNN algorithm.
- Understand the working of the Decision tree and Random forest algorithm.
- Learn the concept of Cross-Validation.

Introduction

In the process of data mining, large data sets are first sorted, then patterns are identified and relationships are established to perform data analysis and solve problems. Attributes represent different features of an object. Different types of attributes are:

- **Binary:** Possesses only two values i.e. True or False

- **Nominal:** When more than two outcomes are possible. It is in Alphabet form rather than being in Integer form.
- **Ordinal:** Values that must have some meaningful order.
- **Continuous:** May have an infinite number of values, it is in float type
- **Discrete:** Finite number of values.

11.1 Supervised Learning

As the name implies, supervised learning involves the involvement of a supervisor who often serves as an instructor. In a nutshell, supervised learning is when we teach or train a computer using well-labeled data. This means that certain information has already been labeled with the correct answer.

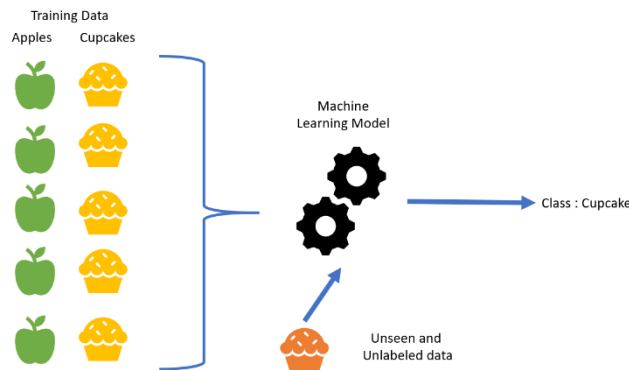


Figure 1: Supervised Learning

Steps Involved in Supervised Learning

- First Determine the type of training dataset
- Collect/Gather the labeled training data.
- Split the training dataset into a training dataset, test dataset, and validation dataset.
- Determine the input features of the training dataset, which should have enough knowledge so that the model can accurately predict the output.
- Determine the suitable algorithm for the model, such as support vector machine, decision tree, etc.
- Execute the algorithm on the training dataset. Sometimes we need validation sets as the control parameters, which are the subset of training datasets.
- Evaluate the accuracy of the model by providing the test set. If the model predicts the correct output, which means our model is accurate.

11.2 Classification

Classification is a data mining function that assigns items in a collection to target categories or classes. The goal of classification is to accurately predict the target class for each case in the data.



You may wish to use classification to predict whether the weather on a particular day will be “sunny”, “rainy” or “cloudy”. Popular classification techniques include decision trees and neural networks.

Classification is a two-step process:

(a) **Learning:** Training data are analyzed by a classification algorithm. Here, the class label attribute is a credit rating, and the learned model or classifier is represented in the form of a classification rule. In the learning step (or training phase), a classification algorithm builds the classifier by analyzing or “learning from” a training set. A tuple, X , is represented by an N -dimensional attribute vector,

$$X = \{x_1, x_2, \dots, x_N\}$$

Each tuple, X , is assumed to belong to a predefined class as determined by another database attribute called the class label attribute. The individual tuples making up the training set are referred to as training tuples and are randomly sampled from the database under analysis.

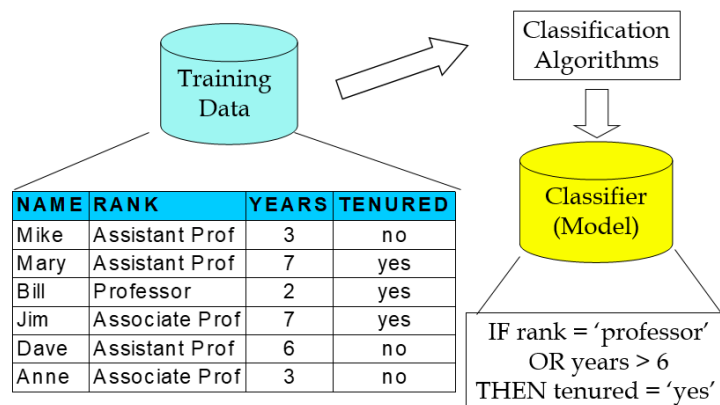


Figure 2: Learning Step

(b) **Classification:** Test data are used to estimate the accuracy of the classification rules. If the accuracy is considered acceptable, the rules can be applied to the classification of new data tuples. The model is used to predict class labels and testing the constructed model on test data and hence estimate the accuracy of the classification rules.

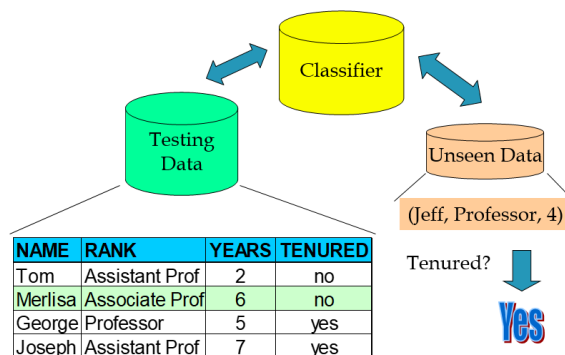


Figure 3: Classification Step

11.3 Issues regarding Classification

To prepare the data for classification and prediction, the following preprocessing steps may be applied to the data to help improve the accuracy, efficiency, and scalability of the classification or prediction process.

1. **Data cleaning:** This refers to the preprocessing of data to remove or reduce noise (by applying smoothing techniques, for example), and the treatment of missing values (e.g., by replacing a missing value with the most commonly occurring value for that attribute, or with the most probable value based on statistics). Although most classification algorithms have some mechanisms for handling noisy or missing data, this step can help reduce confusion during learning.
2. **Relevance analysis:** Many of the attributes in the data may be irrelevant to the classification or prediction task. For example, data recording the day of the week on which a bank loan application was filled is unlikely to be relevant to the success of the application. Furthermore, other attributes may be redundant. Hence, relevance analysis may be performed on the data to remove any irrelevant or redundant attributes from the learning process. In machine learning, this step is known as feature selection. Including such attributes may otherwise slow down, and possibly mislead the learning step. Ideally,

the time spent on relevance analysis, when added to the time spent on learning from the resulting “reduced” feature subset, should be less than the time that would have been spent on learning from the original set of features. Hence, such analysis can help improve classification efficiency and scalability.

3. **Data Transformation and reduction: Normalization** involves scaling all values for a given attribute to make them fall within a small specified range. It involves scaling all values for a given attribute so that they fall within a small specified range, such as -1.0 to 1.0, or 0 to 1.0. In methods that use distance measurements, for example, this would prevent attributes with initially large ranges (like, say income) from outweighing attributes with initially smaller ranges (such as binary attributes).
4. **Generalization:** The data can also be transformed by generalizing it to the higher concept. For this purpose, we can use the concept hierarchies. This is particularly useful for continuous-valued attributes. For example, numeric values for the attribute income may be generalized to discrete ranges such as low, medium, and high. Similarly, nominal-valued attributes, like streets, can be generalized to higher-level concepts, like a city. Since generalization compresses the original training data, fewer input/output operations may be involved during learning.

11.4 Types of Classifiers

Classifiers can be categorized into two major types:

- Discriminative
- Generative

Discriminative

It is a very basic classifier and determines just one class for each row of data. It tries to model just by depending on the observed data, depends heavily on the quality of data rather than on distributions.



Logistic Regression

Suppose there are few students and the Result of them are as follows :

Student 1 : Test Score: 9/10, Grades: 8/10 Result: Accepted

Student 2 : Test Score: 3/10, Grades: 4/10, Result: Rejected

Student 3: Test Score: 7/10, Grades: 6/10, Result: to be tested

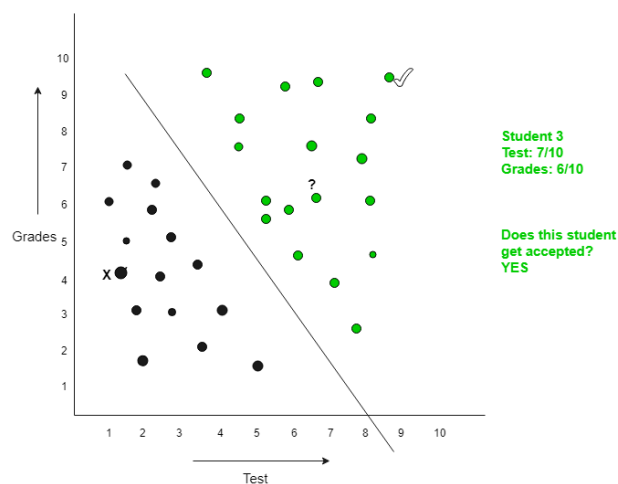


Figure 4: Discriminative Classifier

Generative

It models the distribution of individual classes and tries to learn the model that generates the data behind the scenes by estimating assumptions and distributions of the model. Used to predict the unseen data.



Naive Bayes Classifier

Detecting Spam emails by looking at the previous data. Suppose 100 emails and that too divided in 1:4 i.e. Class A: 25%(Spam emails) and Class B: 75%(Non-Spam emails). Now if a user wants to check that if an email contains the word cheap, then that may be termed as Spam. So, if the email contains the word cheap, what is the probability of it being spam ?? (= 80%).

11.5 Binary Classification

Binary classification is the task of classifying the elements of a set into two groups based on a classification rule. Typical binary classification problems include:

- Medical testing to determine if a patient has a certain disease or not;
- Quality control in industry, deciding whether a specification has been met;
- In information retrieval, deciding whether a page should be in the result set of a search or not.

Typically, binary classification tasks involve one class that is the normal state, and another class that is the abnormal state.



“cancer not detected” is the normal state of a task that involves a medical test and “cancer detected” is the abnormal state.

The class for the normal state is assigned the class label 0 and the class with the abnormal state is assigned the class label 1.

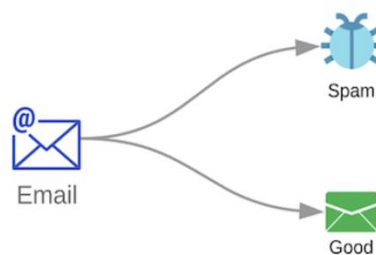


Figure 5: Binary Classification.

Popular algorithms that can be used for binary classification include:

- Logistic Regression
- k-Nearest Neighbors
- Decision Trees
- Support Vector Machine
- Naive Bayes

11.6 Why Bayesian Classification?

It is a statistical classifier, it can predict class membership probabilities such as the probability that a given tuple belongs to a particular class. It is based on Bayes' Theorem. A simple Bayesian classifier, naïve Bayesian classifier, has comparable performance with decision tree and selected neural network classifiers

Bayesian Theorem: Basics

Let X be a data sample ("*evidence*"): class label is unknown. Let H be a *hypothesis* that X belongs to class C . Classification is to determine $P(H|X)$ is the posteriori probability, of H , conditioned on X .

- X is a 35-year-old customer with an income of \$40,000.

Then $P(H|X)$ reflects the probability that customer X will buy a computer given that we know the customer's age and income. In contrast, $P(H)$ is the prior probability, of H . For our example, this is the probability that any given customer will buy a computer, regardless of age, income, or any other information. $P(H)$, which is independent of X . $P(X|H)$ (*posteriori probability*), the probability of X conditioned on H . That is, it is the probability that a customer, X , is 35 years old and earns \$40,000, given that we know the customer will buy a computer.

$P(X)$ is the prior probability of X . Using our example, it is the probability that a person from our set of customers is 35 years old and earns \$40,000.

Bayesian Theorem

Given training data X , posteriori probability of a hypothesis H , $P(H|X)$, follows the Bayes theorem

$$P(H|X) = \frac{P(X|H)P(H)}{P(X)}$$

Derivation of Naïve Bayesian Classifier

Let D be a training set of tuples and their associated class labels, and each tuple is represented by an n -D attribute vector $X = (x_1, x_2, \dots, x_n)$. Suppose there are m classes $C_1, C_2, \dots, C_m$. Classification is to derive the maximum posteriori, i.e., the maximal $P(C_i|X)$. This can be derived from Bayes' theorem

$$P(C_i|X) = \frac{P(X|C_i)P(C_i)}{P(X)}$$

$$P(C_i|X) = P(X|C_i)P(C_i)$$

Since $P(X)$ is constant for all classes, only needs to be maximized. A simplified assumption: attributes are conditionally independent (i.e., no dependence relation between attributes):

$$P(X|C_i) = \prod_{k=1}^n P(x_k | C_i) = P(x_1 | C_i) \times P(x_2 | C_i) \times \dots \times P(x_n | C_i)$$

Example

For instance, to compute $P(X|C_i)$, we consider the following:

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Figure 6: Data Set

Class:

C1:buys_computer = 'yes'

C2:buys_computer = 'no'

Data sample

$X = (\text{age} \leq 30, \text{Income} = \text{medium}, \text{Student} = \text{yes}, \text{Credit_rating} = \text{Fair}).$

$P(C_i): P(\text{buys_computer} = \text{"yes"}) = 9/14 = 0.643$

$P(\text{buys_computer} = \text{"no"}) = 5/14 = 0.357$

Compute $P(X | C_i)$ for each class

$P(\text{age} = \text{"<=30"} | \text{buys_computer} = \text{"yes"}) = 2/9 = 0.222$

$P(\text{age} = \text{"<= 30"} | \text{buys_computer} = \text{"no"}) = 3/5 = 0.6$

$P(\text{income} = \text{"medium"} | \text{buys_computer} = \text{"yes"}) = 4/9 = 0.444$

$P(\text{income} = \text{"medium"} | \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$

$P(\text{student} = \text{"yes"} | \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$

$P(\text{student} = \text{"yes"} | \text{buys_computer} = \text{"no"}) = 1/5 = 0.2$

$P(\text{credit_rating} = \text{"fair"} | \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$

$P(\text{credit_rating} = \text{"fair"} | \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$

$X = (\text{age} \leq 30, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit_rating} = \text{fair})$

$P(X | C_i): P(X | \text{buys_computer} = \text{"yes"}) = 0.222 \times 0.444 \times 0.667 \times 0.667 = 0.044$

$P(X | \text{buys_computer} = \text{"no"}) = 0.6 \times 0.4 \times 0.2 \times 0.4 = 0.019$

$P(X | C_i) \cdot P(C_i): P(X | \text{buys_computer} = \text{"yes"}) \cdot P(\text{buys_computer} = \text{"yes"}) = 0.028$

$P(X | \text{buys_computer} = \text{"no"}) \cdot P(\text{buys_computer} = \text{"no"}) = 0.007$

Therefore, X belongs to class ("buys_computer = yes")

Advantages

- It is simple and easy to implement
- It doesn't require as much training data
- It handles both continuous and discrete data

- It is highly scalable with the number of predictors and data points
- It is fast and can be used to make real-time predictions
- It is not sensitive to irrelevant features

Disadvantages

- Assumption: class conditional independence, therefore loss of accuracy
- Practically, dependencies exist among variables
E.g., hospitals: patients: Profile: age, family history, etc.
Symptoms: fever, cough, etc., Disease: lung cancer, diabetes, etc.
- Dependencies among these cannot be modeled by Naïve Bayesian Classifier.

How effective are Bayesian Classifiers?

In theory, Bayesian classifiers have the minimum error rate in comparison to all other classifiers. However, in practice, this is not always the case owing to inaccuracies in the assumptions made for its use, such as class conditional independence, and the lack of available probability data. However, various empirical studies of this classifier in comparison to the decision tree and neural network classifiers have found it to be comparable in some domains.

Bayesian classifiers are also useful in that they provide a theoretical justification for other classifiers which do not explicitly use Bayes theorem. For example, under certain assumptions, it can be shown that many neural network and curve fitting algorithms output the maximum posteriori hypothesis, as does the naive Bayesian classifier.



“Classification is a data mining technique used to predict group membership for data instances”. Discuss.

11.7 Association based classification

Association rule mining finds interesting associations and relationships among large sets of data items. It shows how frequently an itemset occurs in a transaction. A typical example is a Market Based Analysis. An associative classifier (AC) is a type of supervised learning model that assigns a target value using association rules.

Market-Based Analysis is one of the key techniques used by large relations to show associations between items. It allows retailers to identify relationships between the items that people buy together frequently.

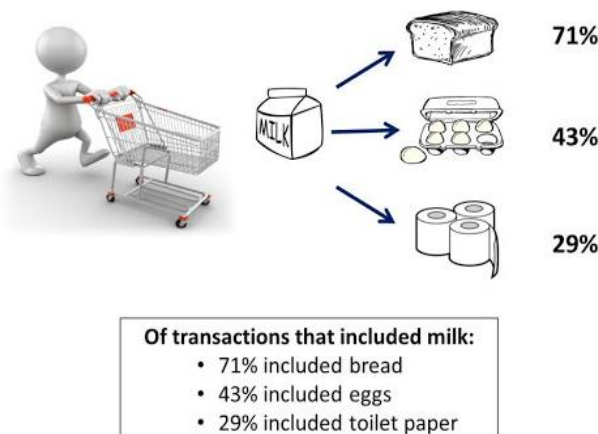


Figure 7: Association based classification

Given a set of transactions, we can find rules that will predict the occurrence of an item based on the occurrences of other items in the transaction.

Table 1: Set of transactions

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Before we start defining the rule, let us first see the basic definitions.

- **Support Count()- σ** Frequency of occurrence of an itemset. Here $\sigma(\{\text{Milk, Bread, Diaper}\})=2$
- **Frequent Itemset** - An itemset whose support is greater than or equal to the minsup threshold.
- **Association Rule**- An implication expression of form $X \rightarrow Y$, where X and Y are any 2 itemsets.



$\{\text{Milk, Diaper}\} \rightarrow \{\text{Beer}\}$

Support(s)

The number of transactions that include items in the {X} and {Y} parts of the rule as a percentage of the total number of transactions. It is a measure of how frequently the collection of items occur together as a percentage of all transactions.

$$\text{Support} = \sigma(X+Y) / \text{total}$$

It is interpreted as a fraction of transactions that contain both X and Y.

Confidence(c)

It is the ratio of the no of transactions that includes all items in {B} as well as the no of transactions that includes all items in {A} to the no of transactions that includes all items in {A}.

$$\text{Conf}(X \Rightarrow Y) = \text{Supp}(X \cup Y) \div \text{Supp}(X)$$

It measures how often each item in Y appears in transactions that contain items in X also.



$\{\text{Milk, Diaper}\} \Rightarrow \{\text{Beer}\}$

$$\begin{aligned} s &= \sigma(\{\text{Milk, Diaper, Beer}\}) \div |T| \\ &= 2/5 \\ &= 0.4 \\ \\ c &= \sigma(\text{Milk, Diaper, Beer}) \div \sigma(\text{Milk, Diaper}) \\ &= 2/3 \\ &= 0.67 \end{aligned}$$

11.8 Rule-Based Classifier

For classification, a rule-based classifier employs a collection of IF-THEN rules. From the following, we can express a rule:

IF condition THEN conclusion

Points to remember:

- The IF part of the rule is referred to as the rule antecedent or precondition.
- The rule's THEN part is referred to as rule consequent.
- The condition's antecedent part consists of one or more attribute checks that are logically ANDed.
- The consequent part consists of class prediction.

Application of Rule-Based Classifier

A rule r covers an instance x if the attributes of the instance satisfy the condition of the rule

R1: (Give Birth = no) ^ (Can Fly = yes) -> Birds

R2: (Give Birth = no) ^ (Live in Water = yes) -> Fishes

R3: (Give Birth = yes) ^ (Blood Type = warm) -> Mammals

R4: (Give Birth = no) ^ (Can Fly = no) -> Reptiles

R5: (Live in Water = sometimes) -> Amphibians

Table 2: Test Dataset

Name	Blood Type	Give Birth	Can Fly	Live in Water	Class
Hawk	warm	no	yes	no	?
grizzly bear	warm	yes	no	no	?

The rule R1 covers a hawk => Bird

The rule R3 covers the grizzly bear => Mammal

How does Rule-based Classifier Work?

R1: (Give Birth = no) ^ Can Fly = yes -> Birds

R2: (Give Birth = no) ^ (Live in Water = yes) -> Fishes

R3: (Give Birth = yes) ^ (Blood Type = warm) -> Mammals

R4: (Give Birth = no) ^ (Can Fly = no) -> Reptiles

R5: (Live in Water = sometimes) -> Amphibians

Name	Blood Type	Give Birth	Can Fly	Live in Water	Class
lemur	warm	yes	no	no	?
turtle	cold	no	no	sometimes	?
dogfish shark	cold	yes	no	yes	?

A lemur triggers rule R3, so it is classified as a mammal A turtle triggers both R4 and R5

A dogfish shark triggers none of the rules

From Decision Trees To Rules

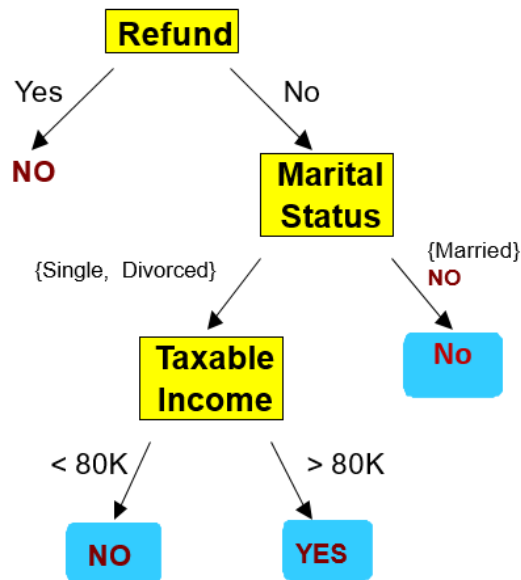


Figure 8: Decision Tree for Rule Generation

The Following are the classification rules generated from the given decision tree.

Classification Rules

(Refund=Yes) ==> No

(Refund=No, Marital Status={Single,Divorced},Taxable Income<80K) ==> No

(Refund=No, Marital Status={Single,Divorced},Taxable Income>80K) ==> Yes

(Refund=No, Marital Status={Married}) ==> No.



Explain how will you remove the training data covered by rule R.

11.9 K-Nearest Neighbour(KNN)

The KNN algorithm is a type of supervised machine learning algorithm that can be used to solve both classification and regression predictive problems. However, in industry, it is primarily used to solve classification and prediction problems. The following two characteristics would be a good way to describe KNN:

1. **Lazy learning algorithm:** KNN is a lazy learning algorithm since it doesn't have a dedicated training process and instead uses all of the data for training and classification.
2. **Non-parametric learning algorithm:** KNN is also a non-parametric learning algorithm since it makes no assumptions about the underlying data.

Working of KNN Algorithm

The KNN algorithm predicts the values of new datapoints using 'feature similarity,' which means that the new data point will be assigned a value based on how closely it matches the points in the training set. With the aid of the steps below, we can understand how it works.

Step1: We must load both training and test data in the first step of KNN.

Step2: The value of K, i.e. the closest data points, must then be chosen. Any integer can be used as K.

Step3: For each point in the test data do the following :

- Calculate the distance between each row of training data and the test data using one of the following methods: Euclidean, Manhattan, or Hamming distance. The Euclidean method is the most widely used method for calculating distance.
- Sort them in ascending order based on the distance value.
- The top K rows of the sorted array will then be chosen.
- The test point will now be assigned a class based on the most common class of these rows.

Example

Customer	Age	Income	No. credit cards	Response
John	35	35K	3	No
Rachel	22	50K	2	Yes
Hannah	63	200K	1	No
Tom	59	170K	1	No
Nellie	25	40K	4	Yes
David	37	50K	2	?

Figure 9: dataset for Finding the nearest neighbor

Table 3: Distance from each customer

Distance from David
John= $\text{sqrt} [(35-37)^2 + (35-50)^2 + (3-2)^2] = 15.16$
Rachel= $\text{sqrt} [(22-37)^2 + (50-50)^2 + (2-2)^2] = 15$
Hannah= $\text{sqrt} [(63-37)^2 + (200-50)^2 + (1-2)^2] = 152.23$
Tom= $\text{sqrt} [(59-37)^2 + (170-50)^2 + (1-2)^2] = 122$
Nellie= $\text{sqrt} [(25-37)^2 + (40-50)^2 + (4-2)^2] = 15.74$

As per the distance calculation, nearest to David is Rachel so the predicted class for David is the same as Rachel.

Table 4: Predicted class for David

Customer	Age	Income (K)	No. cards	Response
----------	-----	------------	-----------	----------

John	35	35	3	No
Rachel	22	50	2	Yes
Hannah	63	200	1	No
Tom	59	170	1	No
Nellie	25	40	4	Yes
David	37	50	2	Yes

11.10 Decision Tree Classifier

A decision tree is a flow-chart-like tree structure, where each internal node denotes a test on an attribute, each branch represents an outcome of the test, and leaf nodes represent classes or class distributions. The topmost node in a tree is the root node. The following decision tree is for the concept of buy computer that indicates whether a customer at a company is likely to buy a computer or not.

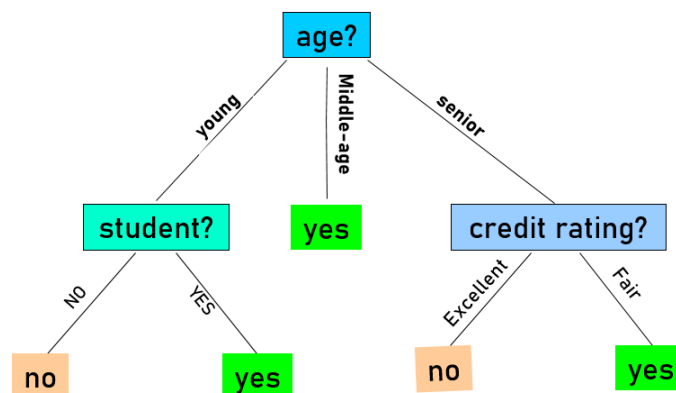


Figure 10: Decision Tree for the concept of buy computer

Decision Tree Induction

The automatic generation of decision rules from examples is known as rule induction or automatic rule induction.

Generating decision rules in the implicit form of a decision tree is also often called rule induction, but the terms tree induction or decision tree inductions are sometimes preferred.

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Figure 11: Dataset for buys_computer

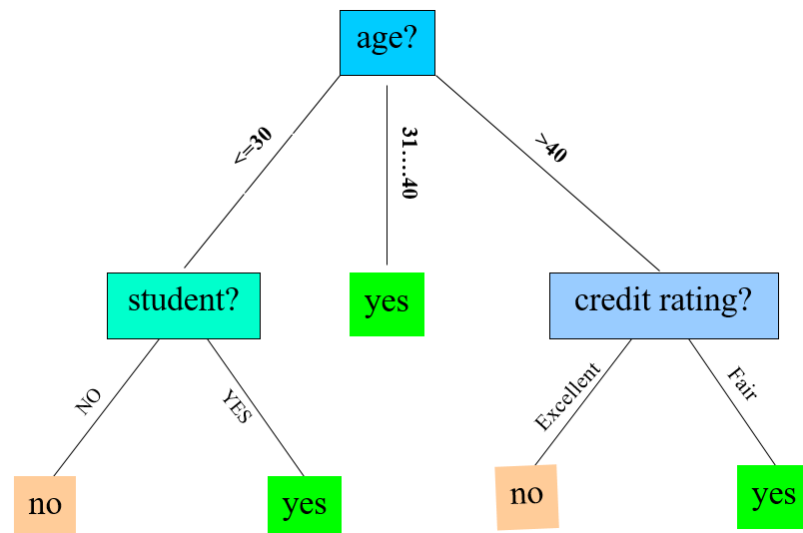


Figure 12: A Decision Tree for "buys_computer"

How are decision trees used for classification?

Given a tuple, X , for which the associated class label is unknown. The attribute values of the tuple are tested against the decision tree. A path is traced from the root to a leaf node, which holds the class prediction for that tuple. Decision trees can easily be converted to classification rules.

Why are decision tree classifiers so popular?

The construction of decision tree classifiers does not require any domain knowledge or parameter setting. Decision trees can handle multidimensional data. Their representation of acquired knowledge in tree form is intuitive and generally easy to understand by humans. The learning and classification steps of decision tree induction are simple and fast.

Attribute Selection Measure

The key problem that emerges when implementing a Decision tree is how to choose the best attribute for the root node and sub-nodes. So, there is a technique called Attribute Selection Measure, or ASM, that can be used to solve such problems. We can easily pick the best attribute for the tree's nodes using this measurement. There are two commonly used ASM techniques are:

- Information Gain

- Gini Index

Information Gain

The calculation of changes in entropy after segmenting a dataset based on an attribute is known as information gain. It determines how much data a function provides about a class. We split the node and built the decision tree based on the importance of information gain. The highest information gain node/attribute is split first in a decision tree algorithm, which always seeks to maximize the amount of information gain. The formula below can be used to measure it.

$$Info_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times Info(D_j)$$

Where

$$Info(D) = -\sum_{i=1}^m p_i \log_2(p_i)$$

In simple terms it can be represented as :

Information Gain= Entropy(S)- [(Weighted Avg) *Entropy(each feature)]

Entropy: Entropy is a metric to measure the impurity in a given attribute. It specifies randomness in data. Entropy can be calculated as:

$$Entropy(s) = -P(yes) \log_2 P(yes) - P(no) \log_2 P(no)$$

Where,

S= Total number of samples

P(yes)= probability of yes

P(no)= probability of no

The information gained by branching on attribute A is represented as:

$$Gain(A) = Info(D) - Info_A(D)$$

Gini Index

The Gini index is a metric of impurity or purity used in the CART (Classification and Regression Tree) algorithm to build a decision tree. In comparison to a high Gini index, an attribute with a low Gini index should be favored. It only produces binary splits, and the CART algorithm creates binary splits using the Gini index. The following formula can be used to measure the Gini index:

$$Gini(D) = 1 - \sum_{i=1}^m p_i^2$$

Example

Table 5: Data for calculating attribute selection measures

RID	age	income	student	Credit_rating	Class:buys_computer
1	youth	high	no	fair	no
2	youth	high	no	excellent	no
3	middle-aged	high	no	fair	yes

4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle-aged	low	yes	excellent	yes
8	youth	medium	no	fair	no
9	youth	low	yes	fair	yes
10	senior	medium	yes	fair	yes
11	youth	medium	yes	excellent	yes
12	middle-aged	medium	no	excellent	yes
13	middle-aged	high	yes	fair	yes
14	senior	medium	no	excellent	no

The class label attribute, *buys_computer*, has two distinct values (namely, {yes, no}); therefore, there are two distinct classes (i.e., $m = 2$). There are nine tuples of class, yes, and five tuples of class no.

$$Info(D) = -\sum_{i=1}^m p_i \log_2(p_i)$$

$$Info(D) = -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) = 0.940 \text{ bits.}$$

We need to look at the distribution of yes and no tuples for each category of age. For the age category “youth,” there are two yes tuples and three no tuples. For the category “middle-aged,” there are four yes tuples and zero no tuples. For the category “senior,” there are three yes tuples and two no tuples.

$$\begin{aligned}
 Info_{age}(D) &= \frac{5}{14} \times \left(-\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} \right) + \frac{4}{14} \times \left(-\frac{4}{4} \log_2 \frac{4}{4} \right) \\
 &\quad + \frac{5}{14} \times \left(-\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} \right) \\
 &= 0.694 \text{ bits.}
 \end{aligned}$$

Hence, the gain in information from such partitioning would be:

$$Gain(age) = Info(D) - Info_{age}(D) = 0.940 - 0.694 = 0.246 \text{ bits.}$$

Similarly, we can compute $Gain(income) = 0.029$ bits, $Gain(student) = 0.151$ bits, and $Gain(credit_rating) = 0.048$ bits. Because age has the highest information gain among the attributes, it is selected as the splitting attribute. Node N is labeled with age.

$$\text{GainRatio}(A) = \frac{\text{Gain}(A)}{\text{SplitInfo}_A(D)}.$$

$$\text{SplitInfo}_A(D) = - \sum_{j=1}^v \frac{|D_j|}{|D|} \times \log_2 \left(\frac{|D_j|}{|D|} \right).$$

A test on income splits the data into three partitions, namely low, medium, and high, containing four, six, and four tuples, respectively.

$$\begin{aligned} \text{SplitInfo}_{\text{income}}(D) &= -\frac{4}{14} \times \log_2 \left(\frac{4}{14} \right) - \frac{6}{14} \times \log_2 \left(\frac{6}{14} \right) - \frac{4}{14} \times \log_2 \left(\frac{4}{14} \right) \\ &= 1.557. \end{aligned}$$

Gain(Income) = 0.029.

Gain Ratio (Income) = 0.029 / 1.557 = 0.019.

The Gini index is used in CART. Using the notation previously described, the Gini index measures the impurity of D, a data partition or set of training tuples, as:

$$\text{Gini}(D) = 1 - \left(\frac{9}{14} \right)^2 - \left(\frac{5}{14} \right)^2 = 0.459.$$

Advantages of the Decision Tree

- It is easy to comprehend because it follows the same steps that a person will take when deciding in the real world.
- It can be extremely helpful in resolving decision-making issues.
- It is beneficial to consider all of the potential solutions to a problem.
- In comparison to other algorithms, data cleaning is not required as much.

Disadvantages of the Decision Tree

- The decision tree is complicated because it has several layers.
- It may have an overfitting problem, which the Random Forest algorithm may solve.
- The computational complexity of the decision tree may increase as more class labels are added.



A company is trying to decide whether to bid for a certain contract or not. They estimate that merely preparing the bid will cost £10,000. If their company bid then they estimate that there is a 50% chance that their bid will be put on the “short-list”, otherwise their bid will be rejected. Once “short-listed” the company will have to supply further detailed information (entailing costs estimated at £5,000). After this stage, their bid will either be accepted or rejected. The company estimate that the labor and material costs associated with the contract are £127,000. They are considering three possible bid prices, namely £155,000, £170,000, and £190,000. They estimate that the probability of these bids being accepted (once they have been short-listed) is 0.90, 0.75, and 0.35 respectively. What should the company do and what is the expected monetary value of your suggested course of action?

11.11 Random Forest

Random Forest is a popular machine learning algorithm that belongs to the supervised learning technique. It is based on the concept of ensemble learning, which is a process of combining multiple classifiers to solve a complex problem and to improve the performance of the model.

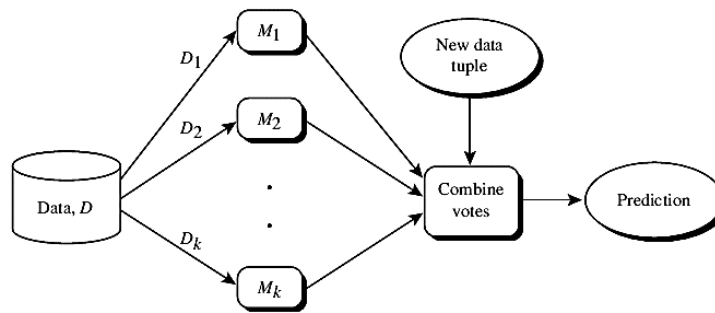


Figure 13: Ensemble Learning

Random Forest is a classifier that contains several decision trees on various subsets of the given dataset and takes the average to improve the predictive accuracy of that dataset. Instead of relying on one decision tree, the random forest takes the prediction from each tree and based on the majority votes of predictions, and predicts the final output. The greater number of trees in the forest leads to higher accuracy and prevents the problem of overfitting.

Working of Random Forest

Step-1: Select random K data points from the training set.

Step-2: Build the decision trees associated with the selected data points (Subsets).

Step-3: Choose the number N for decision trees that you want to build.

Step-4: Repeat Step 1 & 2.

Step-5: For new data points, find the predictions of each decision tree, and assign the new data points to the category that wins the majority votes.

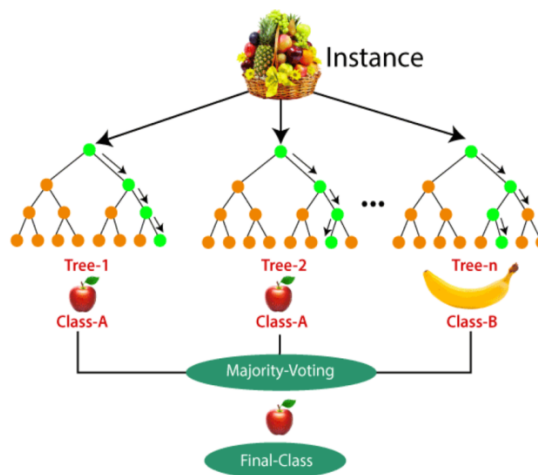


Figure 14: Random Forest

Advantages of Random Forest

- Random Forest is capable of performing both Classification and Regression tasks.
- It is capable of handling large datasets with high dimensionality.
- It enhances the accuracy of the model and prevents the overfitting issue.

Disadvantages of Random Forest

- Complexity is the main disadvantage of Random forest algorithms.
- Construction of Random forests is much harder and time-consuming than decision trees.
- More computational resources are required to implement the Random Forest algorithm.

11.12 Multi-category Classification

The problem of classifying instances into one of three or more classes is known as multiclass or multinomial classification.

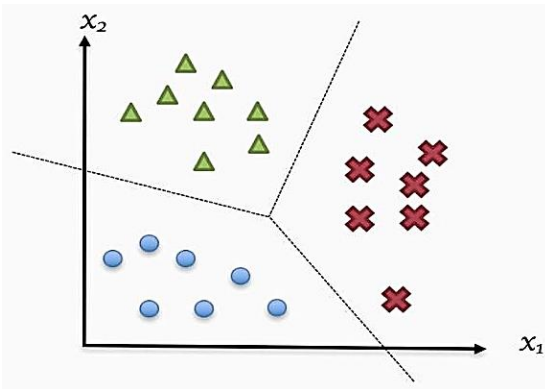


Figure 15: Multi-Category Classification

As a model prediction, multi-class classification is a classification technique that allows us to categorize test data into multiple class labels found in trained data. Multi-class classification methods can be divided into two categories:

- One-versus-all
- One-versus-one

One-verses-all

The one-versus-all method is usually implemented using a “Winner-Takes-All” (WTA) strategy. It constructs M binary classifier models where M is the number of classes. The i^{th} binary classifier is trained with all the examples from i^{th} class W_i with positive labels (typically +1), and the examples from all other classes with negative labels (typically -1).

One-verses-one

The one-versus-one method is usually implemented using a “Max-Wins” voting (MWV) strategy. This method constructs one binary classifier for every pair of distinct classes and so, all together it constructs $M(M - 1)/2$ binary classifiers. The binary classifier C_{ij} is trained with examples from i^{th} class W_i and j^{th} class W_j only, where examples from class W_i take positive labels while examples from class W_j take negative labels.

Some researchers also proposed “all-together” approaches that solve the multi-category classification problem in one step by considering all the examples from all classes together at once. However, the training speed of “all-together” methods is usually slow.

11.13 Cross-Validation

Cross-validation is a technique in which we train our model using the subset of the data-set and then evaluate using the complementary subset of the data-set. The following are the three steps involved in cross-validation:

1. A part of the sample data set should be set aside.
2. Train the model with the rest of the data.
3. Use the data-reserve set portion to test the model.

Methods of Cross-Validation

Validation

In this process, 50% of the data set is used for preparation, and the other 50% is used for testing. The biggest disadvantage of this approach is that we only train on 50% of the dataset; the remaining 50% of the data likely includes crucial details that we're leaving when training our model, resulting in higher bias.

LOOCV (Leave One Out Cross Validation)

This approach iterates for each datapoint by training on the entire data-set but leaving just one data-point from the available data-set. It has some benefits as well as some drawbacks.

Advantage

This technique has the advantage of making use of all data points, resulting in low bias.

Disadvantages

Since we are measuring against a single data point, this approach has the main disadvantage of causing more variance in the testing model. If a data point is an outlier, the variance would be higher. Another disadvantage is that it takes a long time to execute because it iterates over the number of data points.

K-Fold Cross-Validation

This approach divides the data set into k subsets (also known as folds), then trains all of the subsets while leaving one (k-1) subset for evaluation of the trained model. We iterate k times with a different subset reserved for testing purposes each time in this process.



The value of k should always be 10 since a lower value of k leads to validation and a higher value of k leads to the LOOCV method.

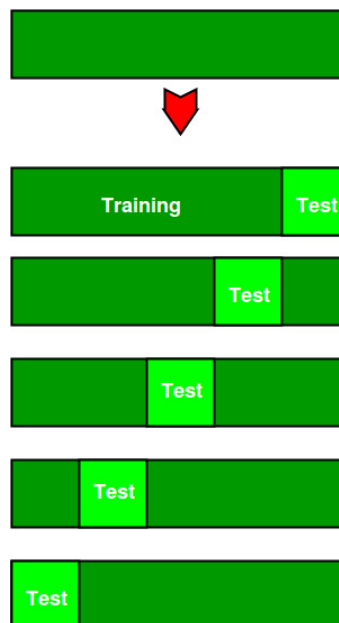


Figure 16: K-Fold cross-validation with K=5

Advantages of a train/test split include:

- Since K-fold cross-validation repeats the train/test split K times, it is K times faster than Leave One Out cross-validation.
- Examining the detailed outcomes of the assessment procedure is easier.

Advantages of Cross-Validation

- Out-of-sample precision can now be estimated more accurately.
- Every observation is used for both training and testing, resulting in a more "effective" use of data.

11.14 Overfitting

When we train a statistical model with a large amount of data (much like fitting ourselves into oversized pants!), it is said to be overfitted. When a model is trained with a large amount of data, it begins to learn from the noise and inaccuracies in the data collection. The model then fails to correctly categorize the data due to too many details and noise. Non-parametric and non-linear

approaches are the causes of overfitting since these types of machine learning algorithms have more flexibility in constructing models based on the dataset and can thus create unrealistic models. If we have linear data, we can use a linear algorithm to avoid overfitting, or we can use decision tree parameters like the maximal depth to avoid overfitting. In a nutshell, overfitting is characterized by a high variance and a low bias.

Bias – A model's assumptions that make a function easier to understand.

Variance: Variance is when you train your data on training data and get a very low error, but when you change the data and then train the same previous model again, you get a very high error.

Summary

- Bayesian classifiers are statistical classifiers. They can predict class membership probabilities, such as the probability that a given sample belongs to a particular class.
- Bayesian classification is based on Bayes theorem. Bayesian classifiers exhibited high accuracy and speed when applied to large databases.
- Classification and prediction methods can be compared and evaluated according to the criteria of Predictive accuracy, Speed, Robustness, Scalability and Interpretability.
- A decision tree is a flow-chart-like tree structure, where each *internal node* denotes a test on an attribute, each *branch* represents an outcome of the test, and *leaf nodes* represent classes or class distributions. The topmost node in a tree is the *root* node.
- The learning of the model is 'supervised' if it is told to which class each training sample belongs. In contrast with unsupervised learning (or clustering), in which the class labels of the training samples are not known, and the number or set of classes to be learned may not be known in advance.
- Naïve Bayes classifiers assume that the effect of a variable value on a given class is independent of the values of other variable. This assumption is called class conditional independence. It is made to simplify the computation and in this sense considered to be "Naïve".
- Non-parametric and non-linear approaches are the causes of overfitting since these types of machine learning algorithms have more flexibility in constructing models based on the dataset and can thus create unrealistic models.
- Cross-validation is a technique in which we train our model using the subset of the data-set and then evaluate using the complementary subset of the data-set.

Keywords

Bayes theorem: Bayesian classification is based on Bayes theorem.

Bayesian belief networks: These are graphical models, which unlike naive Bayesian classifiers, allow the representation of dependencies among subsets of attributes

Bayesian classification: Bayesian classifiers are statistical classifiers.

Classification: Classification is a data mining technique used to predict group membership for data instances.

Decision Tree: A decision tree is a flow-chart-like tree structure, where each *internal node* denotes a test on an attribute, each *branch* represents an outcome of the test, and *leaf nodes* represent classes or class distributions. The topmost node in a tree is the *root* node.

Decision tree induction: The automatic generation of decision rules from examples is known as *rule induction* or *automatic rule induction*.

Overfitting: Decision trees that are too large are susceptible to a phenomenon called as overfitting.

Prediction: Prediction is similar to classification, except that for prediction, the results lie in the future.

Supervised learning: The learning of the model is 'supervised' if it is told to which class each training sample belongs.

Self Assessment

1. Bayesian classifiers are classifiers.
2. Bayesian classification is based on theorem.
3. The learning of the model is if it is told to which class each training sample belongs.
4. A is a flow-chart-like tree structure.
5. The measure is used to select the test attribute at each node in the tree.
6. Which of the following statement is true about the classification?
 - A. It is a measure of accuracy
 - B. It is a subdivision of a set
 - C. It is the task of assigning a classification
 - D. None of the above
7. Data Mining System Classification consists of?
 - A. Database Technology
 - B. Machine Learning
 - C. Information Science
 - D. All of the above
8. A rule-based system consists of a bunch of IF-THEN rules.
 - A. True
 - B. False
 - C. Can't Say
 - D. May be
9. Instead of representing knowledge in a relatively declarative, static way rule-based system represent knowledge in terms of _____ that tell you what you should do or what you could conclude in different situations.
 - A. Raw Text
 - B. A bunch of rules
 - C. Summarized Text
 - D. Collection of various Texts
10. Which of the following distance metric can not be used in k-NN?
 - A. Manhattan
 - B. Minkowski
 - C. Tanimoto
 - D. All can be used
11. Which of the following option is true about k-NN algorithm?
 - A. It can be used for classification
 - B. It can be used for regression
 - C. It can be used in both classification and regression
 - D. None
12. Which of the following will be true about k in k-NN in terms of Bias?
 - A. When you increase the k the bias will be increases
 - B. When you decrease the k the bias will be increases
 - C. Can't say

D. None of these

13. A _____ is a decision support tool that uses a tree-like graph or model of decisions and their possible consequences, including chance event outcomes, resource costs, and utility.

A. Decision tree

B. Graphs

C. Trees

D. Neural Networks

14. What is Decision Tree?

A. Flow-Chart

B. Structure in which internal node represents test on an attribute, each branch represents outcome of test and each leaf node represents class label

C. Flow-Chart & Structure in which internal node represents test on an attribute, each branch represents outcome of test and each leaf node represents class label

D. None of the mentioned

15. Decision Nodes are represented by _____

A. Disks

B. Squares

C. Circles

D. Triangles

Review Questions

Q1. What do you mean by classification in data mining? Write down the applications of classification in business.

Q2. Discuss the issues regarding the classification.

Q3. What is a decision tree? Explain with the help of a suitable example.

Q4. Write down the basic algorithm for decision learning trees.

Q5. Write short notes on the followings:

(a) Bayesian classification

(b) Bayes theorem

(c) Naive Bayesian classification

Q6. Elucidate the usage of cross validation along with its various methods.

Q7. With example explain the KNN algorithm.

Answers for Self Assessment

1. Statistical 2. Bayes 3. Supervised 4. Decision Tree 5. Information gain

6. B 7. D 8. A 9. B 10. D

11. C 12. A 13. A 14. C 15. B

Further Reading

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